

UDAY SUNIL YESHI

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Education

Sardar Patel Institute of Technology, Mumbai

2023 - 2026

Bachelor of Technology in Electronics and Telecommunications, Pursuing 3rd Year, Current CGPA: 8.79

Mumbai

Government Polytechnic Nashik, Nashik

2020 - 2023

Diploma in Electronics and Telecommunications, Percentage: 87.94%

Nashik, Maharashtra

Experience

Intern – Maker's Bench

June 2023

Embedded Systems & Automation Intern

- Developed real-time embedded applications and optimized transistor-based circuits for automation systems.
- Built an autonomous solar panel cleaning bot with ATmega2560, integrating multi-axis control, motor actuation, and remote operation via CT6B protocol.

Research Intern – HMTD Engineering Pvt. Ltd. (SPIT)

Apr 2025 – July 2025

Power Systems & Firmware Development

- Working on firmware and control logic for a DC-DC Converter project focused on precision power control and efficient energy regulation.
- Engaged in low-level hardware-software integration, emphasizing system responsiveness and modular firmware architecture.

Projects

Isolated DC-DC Converter – IEEE IFEC 2025 | *Real-Time Control, Embedded Systems, Power Electronics* Ongoing

- Designing system-level firmware and data interfaces using TI C2000 series to enable fast, stable, and efficient power conversion in a research-grade environment.
- Emphasizing control accuracy, signal stability, and real-time telemetry — core principles in fintech-grade system responsiveness.

Digital PID Control on FPGA | *VLSI, VHDL*

May 2024

- Engineered FPGA-based control system using Verilog/VHDL.
- Tuned system for real-time analytics and sensor feedback.

Smart Prosthetic Hand | *EMG, Arduino Nano*

March 2023

- 3D-printed prosthetic using EMG signals for motor control.
- Implemented signal classification and adaptive actuation.

IoT-Based Smart Heater System | *Real-Time Monitoring, ESP32, Control Analytics*

October 2024

- Created a full-stack IoT system using ESP32 and web interface to control and monitor heater operations based on sensor feedback and user-defined schedules.
- Enabled data logging, trend-based temperature control, and user analytics — simulating smart decision-making frameworks applicable to fintech applications.

Technical Skills

Programming & Scripting: Python, SQL, Java, C, C++, Embedded C, Assembly

Analytics & Simulation Tools: R, MATLAB/Simulink, Scilab, Vivado, Excel

Development Platforms: CCS, VS Code, Thonny IDE, Arduino IDE, Altium Designer, KiCad

Hardware & Prototyping: TI C2000 Series, FPGA (PYNQ-Z2, Artix-7), ESP, Raspberry Pi, Arduino

Extracurricular

BugBuster 2.0 – Hardware Hackathon (1st Place)

Feb 2025

- Led diagnostics and circuit debugging in a global 24hr contest.

VLSI Program – Ingot Semi

June 2024

- Completed RTL to GDSII flow; prototyped on PYNQ-Z2 FPGA.

Self-Learning & Fintech Exploration

Ongoing

- Currently exploring the intersection of technology and finance — actively learning about fintech trends, banking analytics, and Python-based stock data analysis.